

Assignment 8

April 7, 2026 1:23 PM

MATH 3042- Assignment 8

Full Name:

Please show all steps similar to the class work

Set:

1. A database administrator claims that the average query execution time on their PostgreSQL database is 45 ms. After a system upgrade, there are concerns the execution time may have changed. The following query times (in ms) were recorded for a random sample. Test the concern at 5% significance level.

42, 48, 39, 51, 44, 47, 43, 50, 41, 46, 38, 52, 45, 49, 40, 53, 44, 47, 42, 48, 39, 51, 43, 46, 41, 50, 44, 48, 42, 47

(a) Do all steps manually. Use Rstudio to find mean, stdev, and the p-value. Draw your p-value on a curve.

(b) Run the appropriate t.test() in RStudio. Include your t and p values here.

$$H_0: \mu = 45$$

from sample:

$$\text{test statistic: } \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$H_a: \mu \neq 45$$

$$\bar{x} = 45.3$$

$$s = 4.17$$

$$n = 30$$

$$t = \frac{45.3 - 45}{\left(\frac{4.17}{\sqrt{30}}\right)} = 0.4377$$

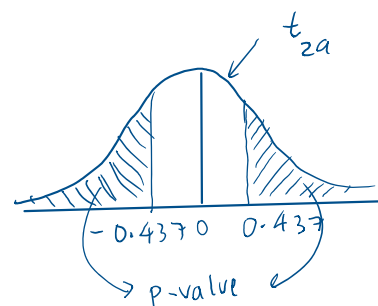
$$p\text{-value} = 2(1 - pt(0.4377, 29)) = 0.6649$$

Decision:

fail to reject H_0

Conclusion: At a 5% significance level the evidence does not suggest that the population mean has Changed.

```
x=
c(42,48,39,51,44,47,43,50,41,46,38,52,45,49,40,53,44,47,42,48,39,51,43,46,41,
50,44,48,42,47)
> t.test(x, mu=45, alternative = "two.sided")
One Sample t-test
data: x
t = 0.43766, df = 29, p-value = 0.6649
alternative hypothesis: true mean is not equal to 45
95 percent confidence interval:
 43.77563 46.89104
sample estimates:
mean of x
45.33333
```



2. A tech company reports that their machine learning model training pipeline has historically taken a mean of 3.2 hours per training run. A data scientist suspects that as the dataset has grown, training time has increased. She randomly selects 25 recent training runs and records their durations (hours):

3.8, 3.1, 4.2, 3.5, 3.9, 3.4, 4.0, 3.7, 3.6, 4.1, 3.3, 3.8, 4.3, 3.5, 3.9, 3.7, 4.1, 3.4, 3.8, 3.6, 4.0, 3.5, 3.9, 3.7, 4.2

- Perform the hypothesis test at $\alpha = 5\%$ level manually. Use Rstudio to find mean, stdev, and the p-value. Draw your p-value on a curve.
- Run the appropriate `t.test()` in RStudio. Include your t and p values here.
- If we change alpha to 1% would our decision change? How?

$$H_0: \mu = 3.2$$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = 9.1245$$

$$H_A: \mu > 3.2 \quad \text{right-tailed test}$$

$$\bar{x} = 3.76$$

$$s = 0.3069$$

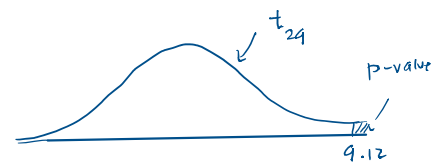
$$n = 25$$

$$p\text{-value} = 1 - pt(9.1245, 24)$$

$$p\text{-value} \approx 0$$

Decision: we reject H_0

Conclusion: The population mean training time has increased.



c) no change. The p-value is still significantly less than α .

3. Historically, 25% of user login attempts to a web application fail due to incorrect credentials. After a UX redesign of the login page, the product team wants to know if the failure rate has changed. In a sample of 300 login attempts, 63 failed.

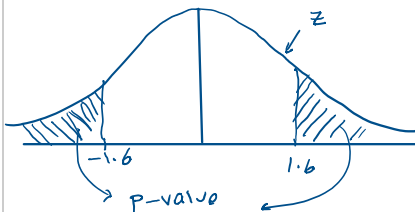
a) Perform the hypothesis test at $\alpha = 5\%$ level manually. Use Rstudio to look up the p-value. Draw your p-value on a curve.

b) Run the appropriate `prop.test()` in RStudio. Include your test statistic and p values here.

$$H_0: \pi = 0.25$$

$$H_A: \pi \neq 0.25$$

Normality check: $300(0.25) \geq 10$ Yes
 $300(0.75) \geq 10$ Yes



$$p\text{-value} = 2 \left(\text{pnorm}(-1.6) \right) \\ = 0.1096$$

$$\hat{p} = \frac{63}{300} = 0.21$$

$$\hat{q} = 1 - 0.21 = 0.79$$

$$\text{test stat: } Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

$$Z = \frac{0.21 - 0.25}{\sqrt{\frac{0.25 \times 0.75}{300}}} = -1.6$$

Decision: the p-value is greater than alpha, we fail to reject H_0

Conclusion: The true percentage of login failures due to incorrect credentials is not significantly different from 25% (doesn't seem to have changed)

```
prop.test(63,300, p=0.25,
alternative = "two.sided")
1-sample proportions test with
continuity correction
data: 63 out of 300, null
probability 0.25
X-squared = 2.3511, df = 1,
p-value = 0.1252
alternative hypothesis: true p
is not equal to 0.25
95 percent confidence
interval:
0.1662081 0.2613875
sample estimates:
p
0.21
```